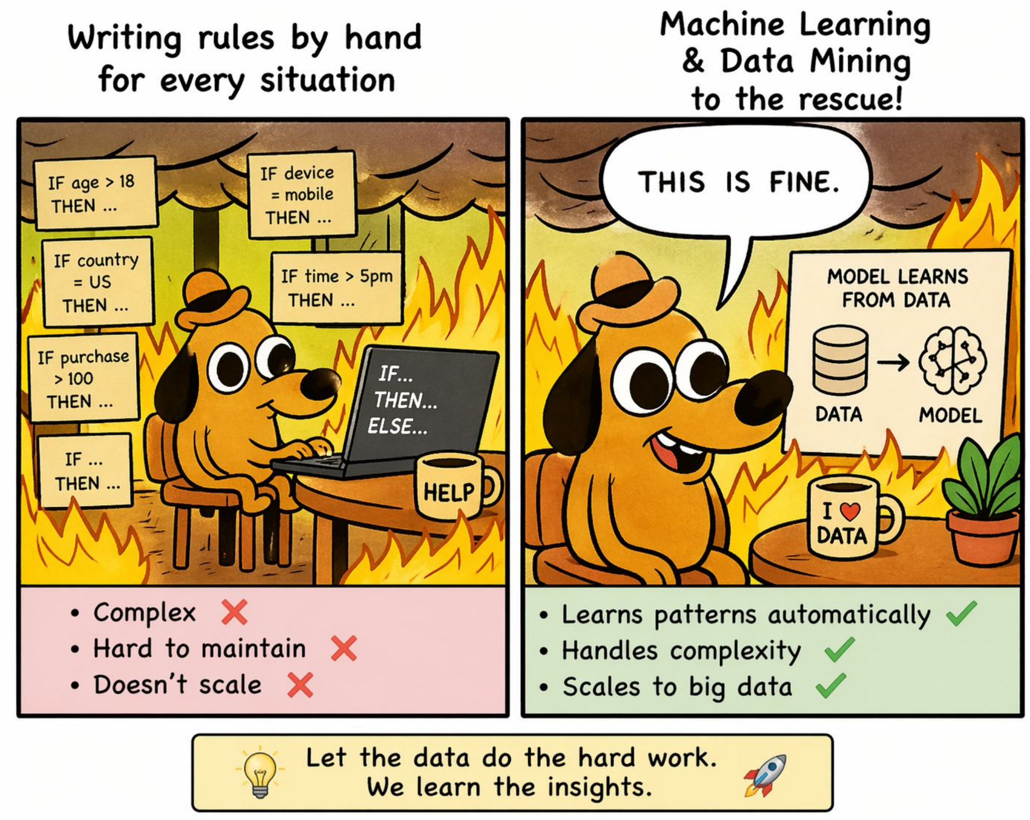
# Introduction to Machine Learning and Data Mining

ICT 3.3 Machine learning and Data Mining

Emanuela Boros, May 2026

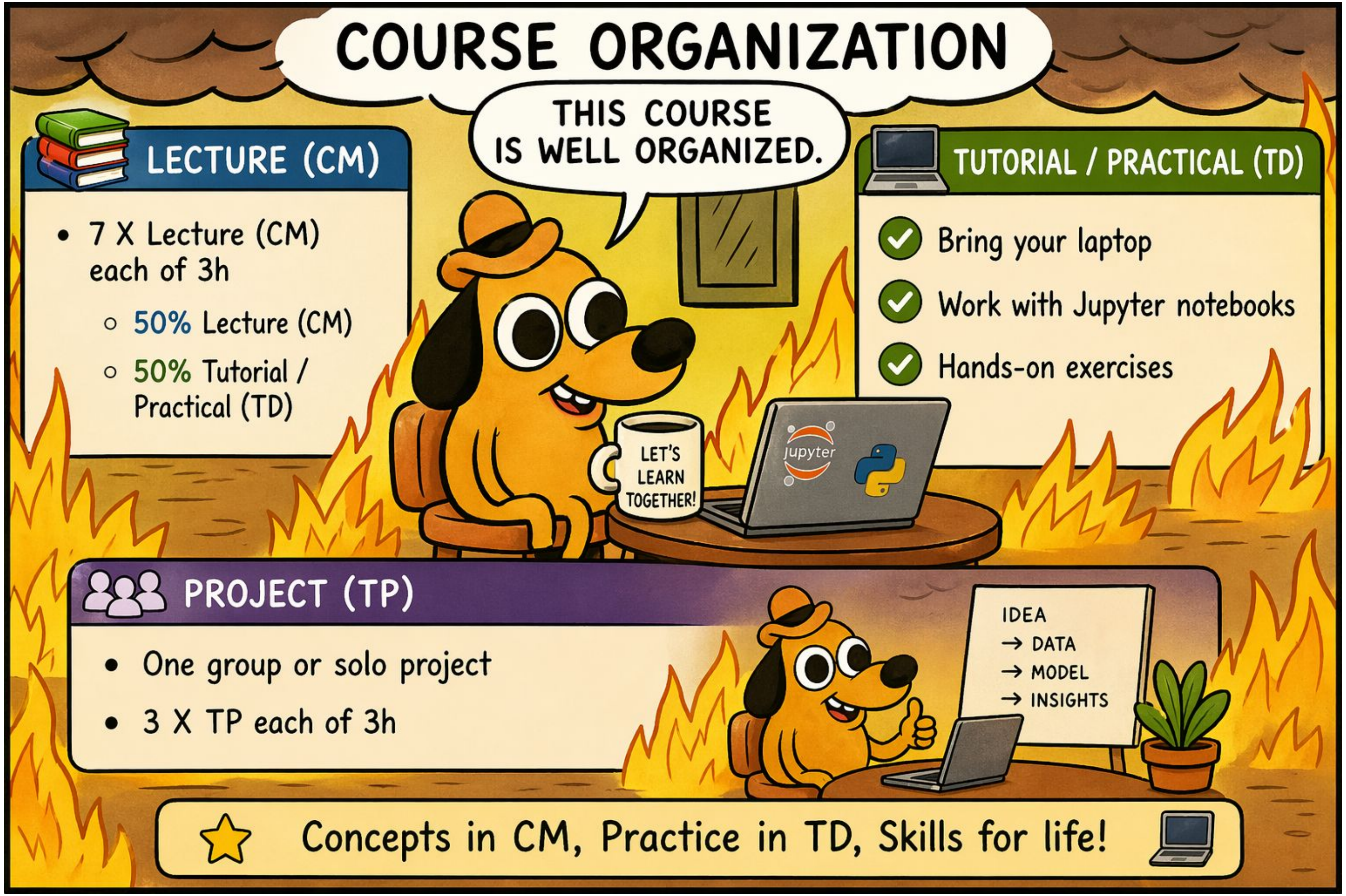
# Welcome

In this course, we'll learn **how systems can learn from data** instead of being explicitly **programmed**.



# By the end of the course, you will:

- Understand core ML paradigms (**supervised** vs **unsupervised**)
- Implement **basic models** from scratch and with libraries
- Evaluate models properly (not just accuracy)
- Understand bias, variance, and overfitting
- Apply ML to small **real-world datasets**
- **Interpret** model outputs

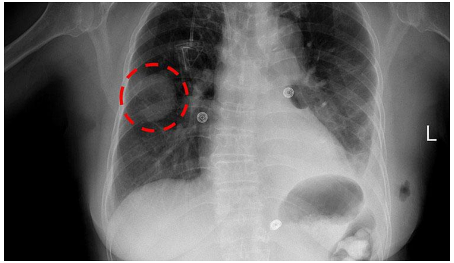

COURSE ORGANIZATION
THIS COURSE IS WELL ORGANIZED.
LECTURE (CM)
• 7 X Lecture (CM) each of 3h
○ 50% Lecture (CM)
○ 50% Tutorial / Practical (TD)
TUTORIAL / PRACTICAL (TD)
Bring your laptop
Work with Jupyter notebooks
Hands-on exercises
LET'S LEARN TOGETHER!
jupyter
PROJECT (TP)
• One group or solo project
• 3 X TP each of 3h
IDEA
→ DATA
→ MODEL
→ INSIGHTS
Concepts in CM, Practice in TD, Skills for life!

# Where do we see Machine Learning?

Can you think of other examples where systems seem to **learn from data**?

# Where do we see Machine Learning?

TESLA





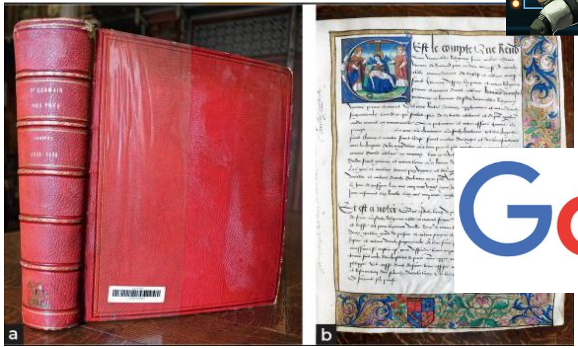
Spotify

Gmail

Siri

ChatGPT

PayPal




Google

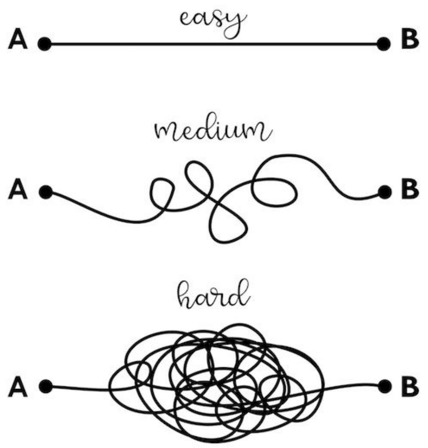
amazon

Google Maps

# What do these systems have in common?

- Data
- Learning
- Patterns

# Why not just write rules?




- Too many cases
- Rules become complex
- Data is messy
- Language is ambiguous: “Apple” → fruit or company?

👉 *Example:* Spam emails

- “FREE!!!” → spam
- but “free event at university” → not spam

# Rules versus Machine Learning

Can you think of another situation where writing rules would be impossible or too complex?

# 🤖 What is Machine Learning?

Machine learning is not that different from how humans learn but it's not magic.

→ *A child learning what a cat looks like* 🐱

- sees many examples of cats
- hears the word "cat" (label)
- learns the pattern

→ does NOT need:

- every colour
- every shape
- every situation

→ ML needs:

- data
- examples
- labels (if supervised)

# 🤖 What is Machine Learning?

Machine learning is the process of learning patterns from data to make predictions or decisions.

“data” → examples (emails, images, text)
“patterns” → relationships (not obvious rules)
“labels” → we know what the data is (what price, what category)
“predictions” → output (spam / not spam, price, category)

# 🤖 What is Machine Learning?

Machine learning is the process of learning patterns from data to make predictions or decisions.

Traditional Programming = Rules + Data → Output

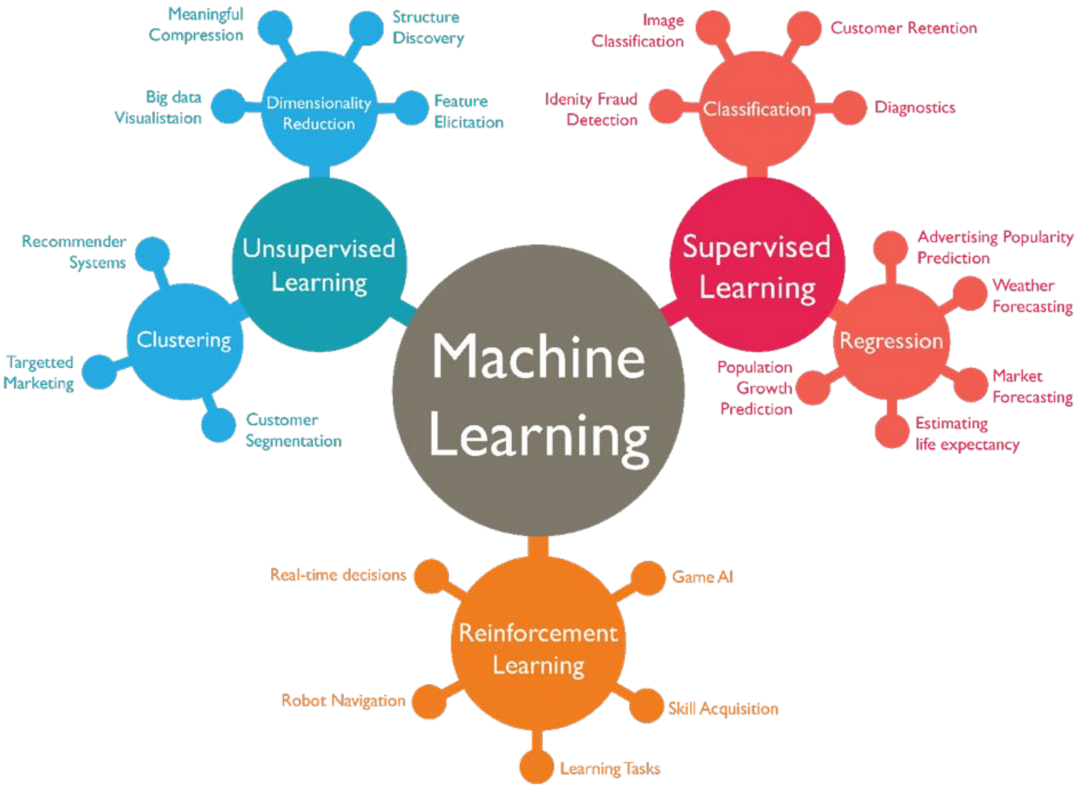
**Machine Learning** = Data + Output → Rules

# Types of Machine Learning

Not all ML problems are the same.

Three main types depending on the availability and types of “labels”:

- Supervised
- Unsupervised
- Reinforcement



# Supervised Learning

We have data and the correct answers.

→ Emails labelled as spam or not spam.
→ Apartments and their prices for a period of time.

# Supervised Learning

Supervised learning is the most common type of Machine Learning.

- Input data (X)
- Correct output (Y)
- Learn a mapping: X → Y

The goal is to learn a function that maps inputs to outputs.” X → Y

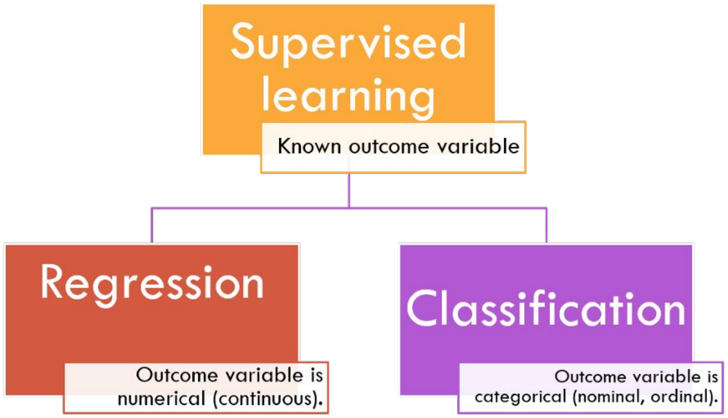
- X → email text
- Y → spam or not spam

It’s called **supervised** because the model is learning from labelled examples → someone already gave the correct answers.

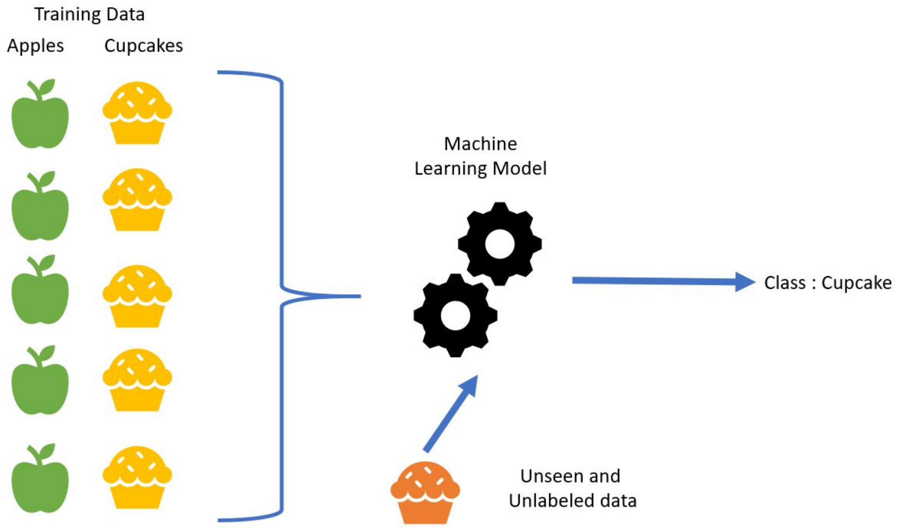
# Supervised Learning: Types

Outputs or “labels” can be:

- categories → classification
- numbers → regression



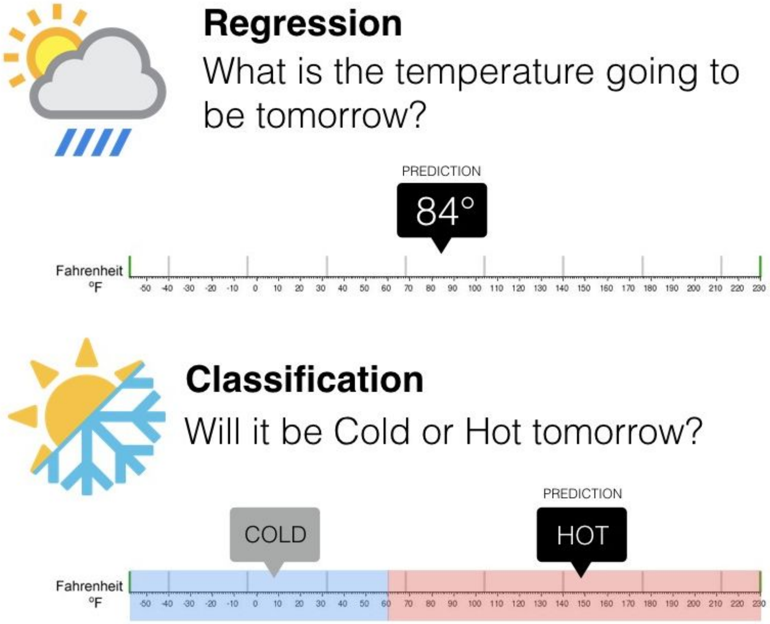
# Supervised Learning: Classification



# Supervised Learning

Is predicting temperature tomorrow classification or regression?

# Supervised Learning



# Supervised Learning

Can you give some examples of classification or regression?

# Supervised Learning: Examples

Classification — Making Decisions 👉 “We predict a category.”

- 📧 Email → Spam / Not Spam
- 🧾 Transaction → Fraud / Legitimate
- 📰 Article → Politics / Sports / Tech

Regression — Predicting Values 👉 “We predict a number.”

- 🏠 Features → House Price
- 🚗 Usage → Car resale value
- 🌡️ Weather data → Temperature tomorrow

# Unsupervised Learning

*What if we have data—but no correct answers?*

- Input data (X)
- No labels (no Y)
- Goal: find structure in data

# Unsupervised Learning

If I give you 1,000 unlabelled images, what could you do with them?

# Unsupervised Learning

Even without labels, there are many useful things we can do.

- Clustering

Group similar data points

- Dimensionality Reduction

Simplify data while keeping important information

- Anomaly Detection

Find unusual or rare cases

# Unsupervised Learning - Clustering

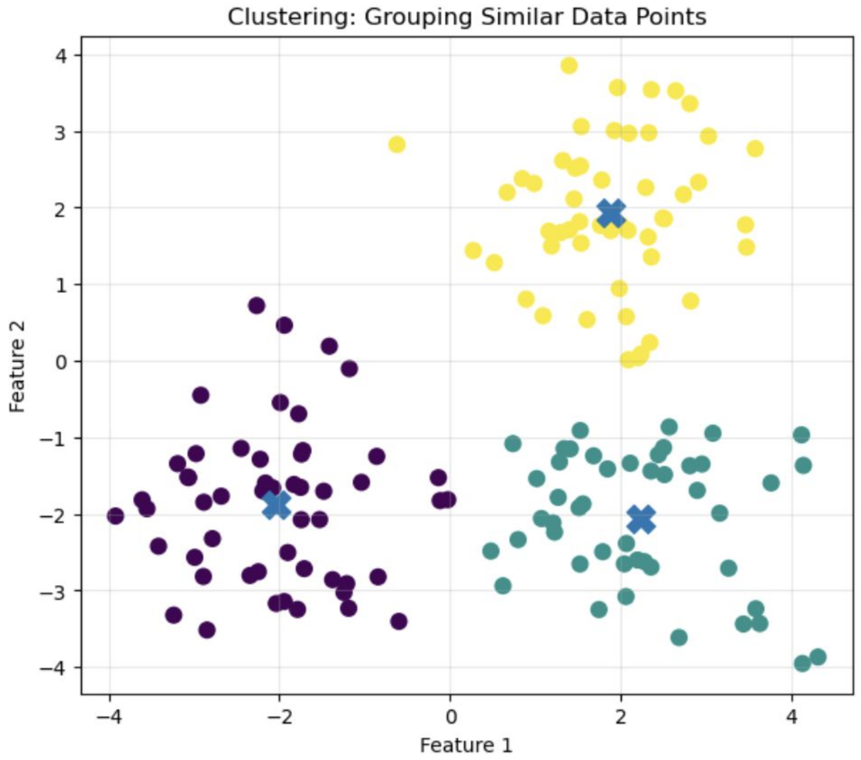
Grouping news articles by topic without knowing the topics in advance.

**Examples**

- group customers
- group images
- group documents

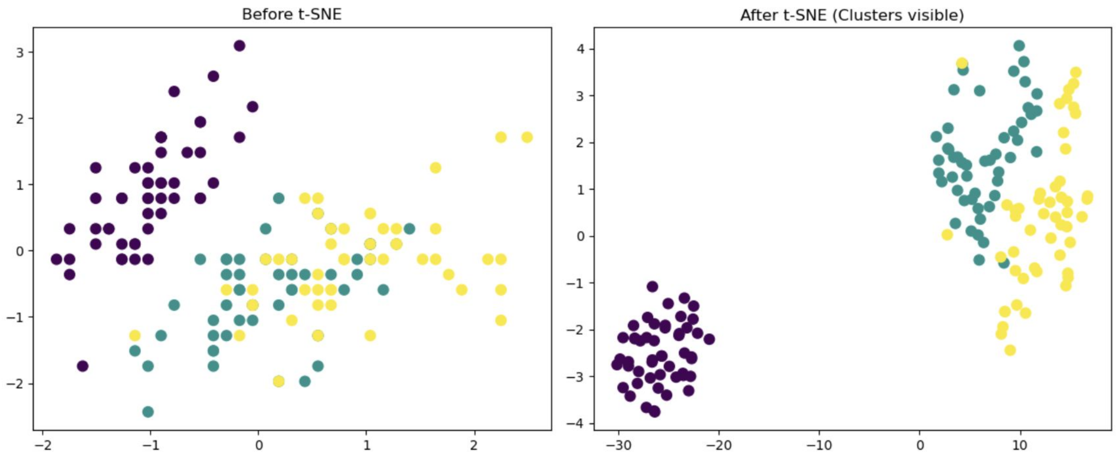
**Methods**

- K-means
- Hierarchical clustering



# Unsupervised Learning - Dimensionality Reduction

Sometimes data is very complex with many features. We reduce it to something simpler while keeping the important structure.

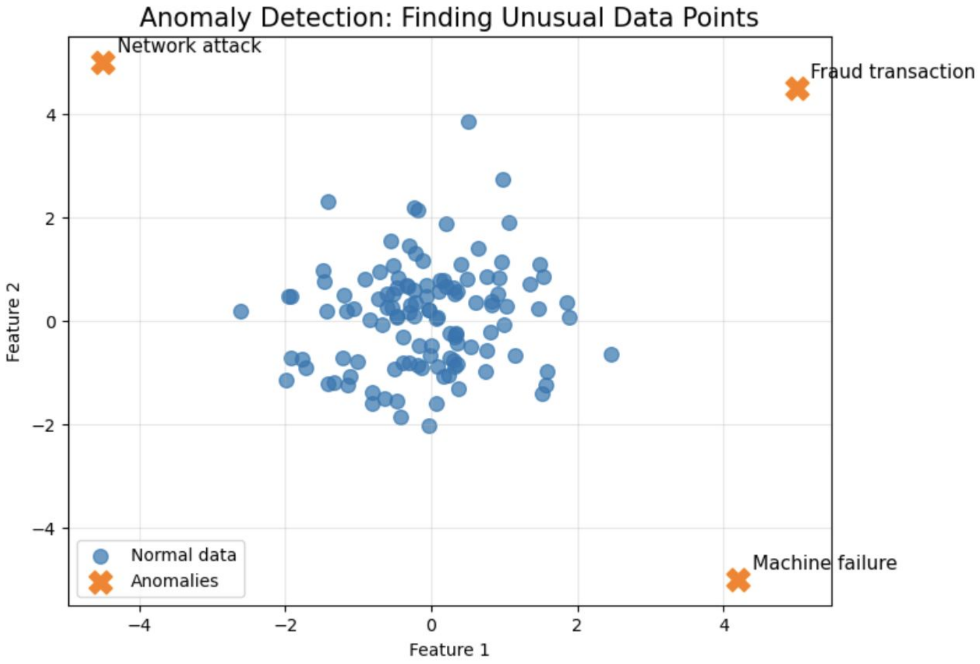


# Unsupervised Learning - Anomaly Detection

We identify data points that are very different from the rest.

Examples:

- fraud detection
- unusual documents



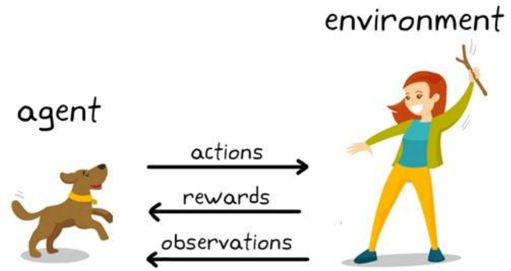
# Unsupervised Learning

Unsupervised learning helps us understand data, not just predict outcomes.

# Unsupervised Learning

Which of these would be useful for exploring a new dataset you don’t understand yet?

# Reinforcement Learning

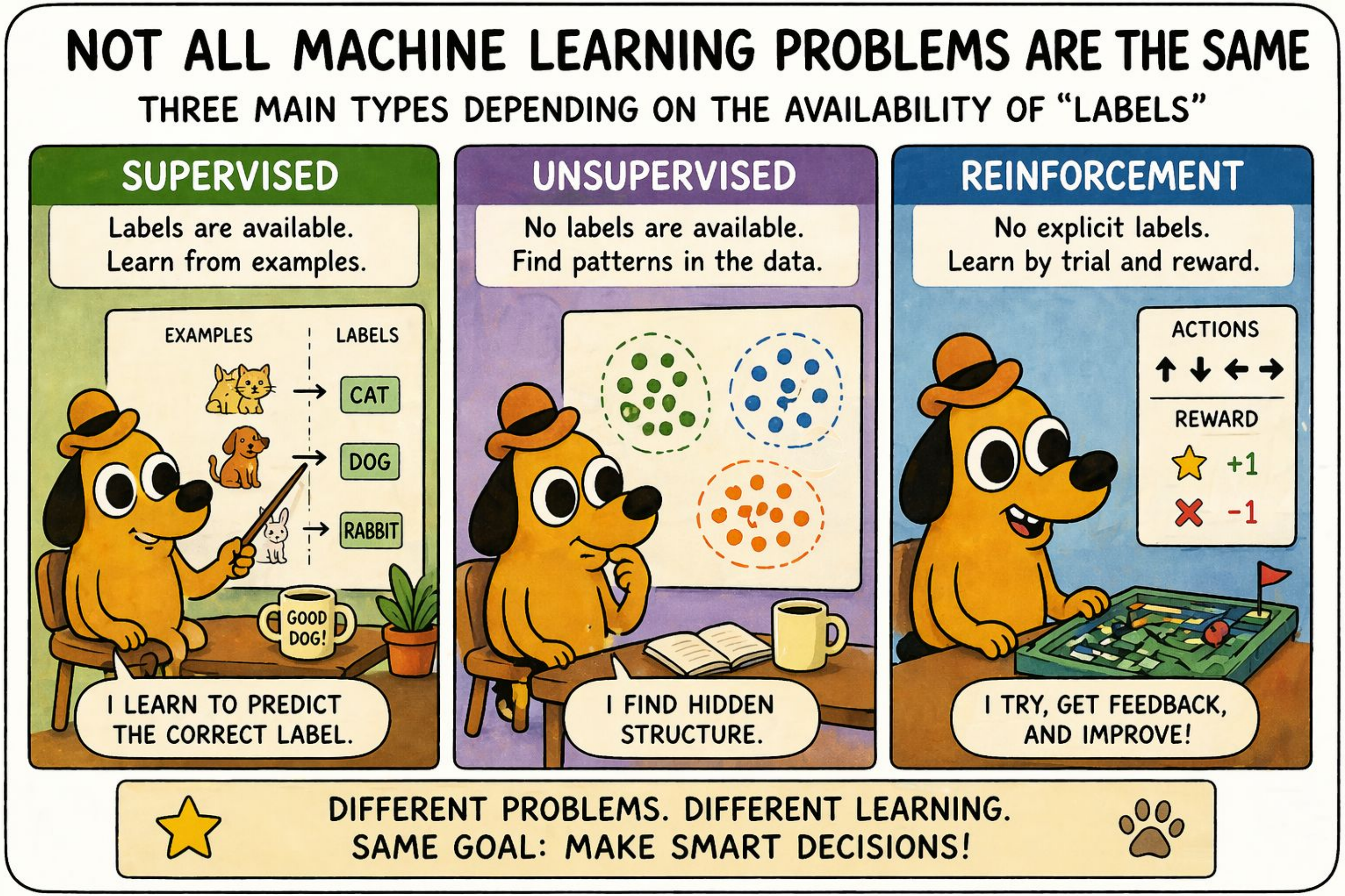



The system learns through interaction and feedback.

Learning by interaction (reward/punishment)

- Agent interacts with environment
- Receives rewards or penalties
- Learns by trial and error

We won't focus on reinforcement learning in this course…but it's important to know it exists.


NOT ALL MACHINE LEARNING PROBLEMS ARE THE SAME
THREE MAIN TYPES DEPENDING ON THE AVAILABILITY OF "LABELS"
SUPERVISED
Labels are available.
Learn from examples.
EXAMPLES
LABELS
CAT
DOG
RABBIT
GOOD DOG!
I LEARN TO PREDICT THE CORRECT LABEL.
UNSUPERVISED
No labels are available.
Find patterns in the data.
I FIND HIDDEN STRUCTURE.
REINFORCEMENT
No explicit labels.
Learn by trial and reward.
ACTIONS
↑ ↓ ← →
REWARD
+1
-1
I TRY, GET FEEDBACK, AND IMPROVE!
DIFFERENT PROBLEMS. DIFFERENT LEARNING.
SAME GOAL: MAKE SMART DECISIONS!

# What is Data Mining?

Data mining is the process of discovering patterns, correlations, and insights in large datasets.

What is a dataset?

# What is a Dataset?

A dataset is a collection of examples used to train a model.

🐶🐱 Dogs vs Cats Dataset

- size
- weight
- ear type
- label (dog/cat)

# What is Data Mining?

Machine learning is about predicting something we don’t know yet.

Machine learning builds a model to predict future outcomes.

- Focus → Prediction
- Future-oriented
- Model-driven

Data mining is about understanding what already exists in the data.

Data mining finds patterns and insights in existing data.

- Focus → Discovery
- Past-oriented
- Insight-driven

# Machine Learning versus Data Mining

Machine Learning = *What will happen?*
Data Mining = *What happened and why?*

So:

Data Mining → finds patterns
Machine Learning → uses patterns to make predictions

# Machine Learning vs Data Mining

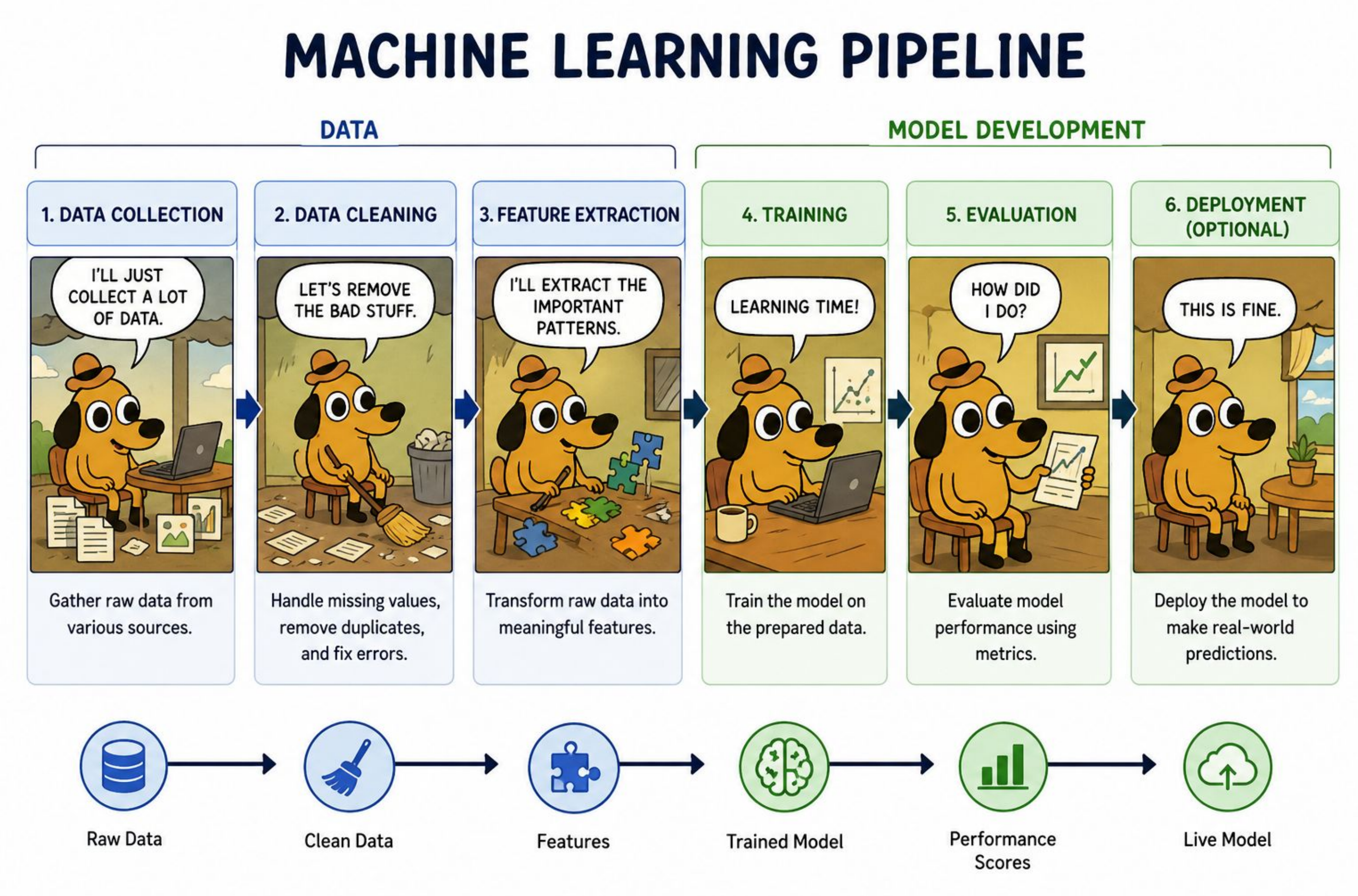
Step 1 — Data Mining: We discover:

- people who buy diapers also buy beer
- people who buy pasta also buy sauce

Step 2 — Machine Learning: Now we use these patterns to predict behaviour.

Example:

- This customer will likely buy beer
- Recommend pasta sauce



MACHINE LEARNING PIPELINE
DATA
MODEL DEVELOPMENT
1. DATA COLLECTION
I'LL JUST COLLECT A LOT OF DATA.
Gather raw data from various sources.
2. DATA CLEANING
LET'S REMOVE THE BAD STUFF.
Handle missing values, remove duplicates, and fix errors.
3. FEATURE EXTRACTION
I'LL EXTRACT THE IMPORTANT PATTERNS.
Transform raw data into meaningful features.
4. TRAINING
LEARNING TIME!
Train the model on the prepared data.
5. EVALUATION
HOW DID I DO?
Evaluate model performance using metrics.
6. DEPLOYMENT (OPTIONAL)
THIS IS FINE.
Deploy the model to make real-world predictions.
Raw Data
Clean Data
Features
Trained Model
Performance Scores
Live Model

# What Makes ML Hard (Reality Check)

This is where you gain credibility.

- Data is messy
- Labels are noisy
- Models are sensitive
- Results can be misleading

Most of ML is not algorithms → it’s dealing with data.

# Data is Messy

- missing values
- duplicates
- inconsistencies

# Labels are Noisy

- humans make mistakes
- ambiguity (especially language)
- disagreement between annotators

# Models are Sensitive

- small changes → different results
- overfitting vs underfitting
- depends on data quality

What happens if the model tries too hard to learn from this messy data?

# Key Challenges in ML

**Overfitting**: Model memorizes instead of learning

**Underfitting**: Model too simple

**Data Quality**: Garbage in → garbage out

# Key Challenges in ML: Overfitting

A dog is usually orange, mid-sized and with an orange hat.

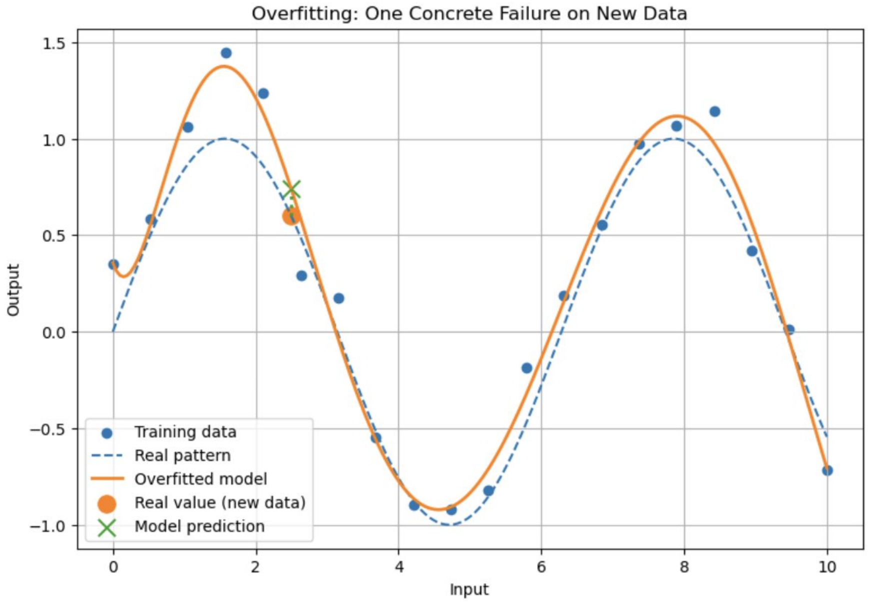
# Key Challenges in ML: Overfitting

I have never seen a grey dog!

# Key Challenges in ML: Overfitting

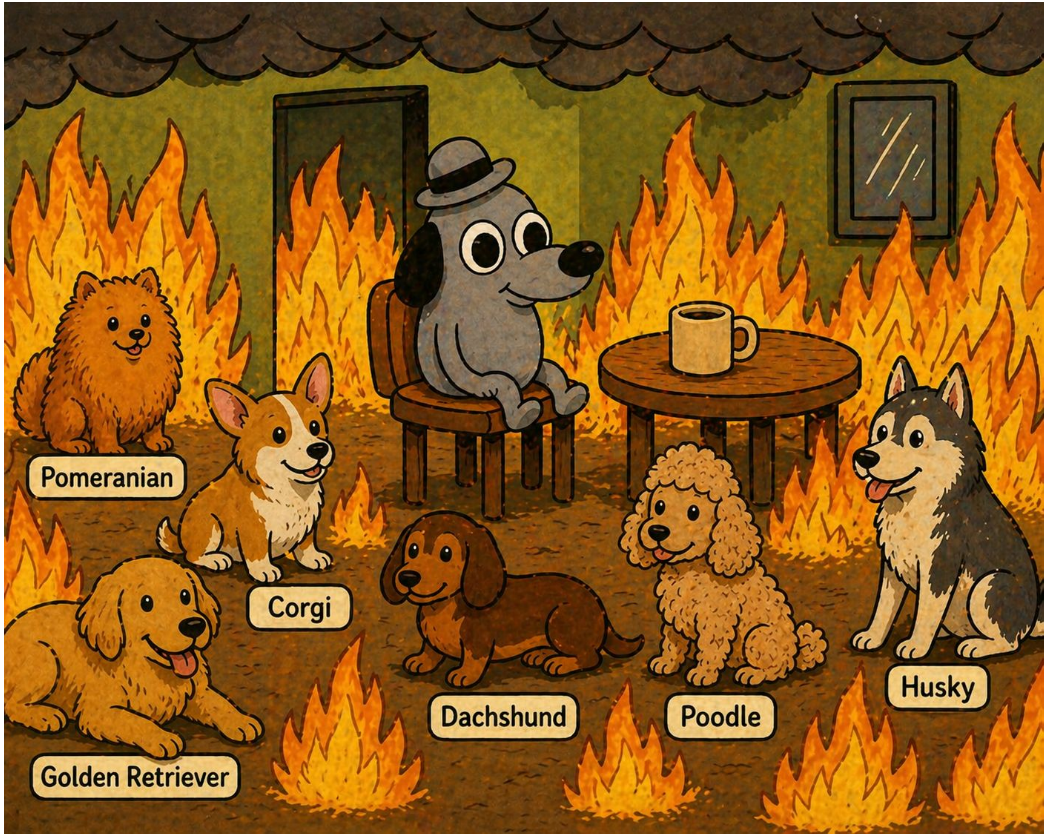
- Model memorizes training data
- Fits noise instead of pattern
- Fails on new data

This model looks perfect… but it's actually bad.



# Key Challenges in ML: Underfitting

All these dogs look the same to me!



# Key Challenges in ML: Underfitting

- Model too simple
- Misses important patterns
- Poor performance

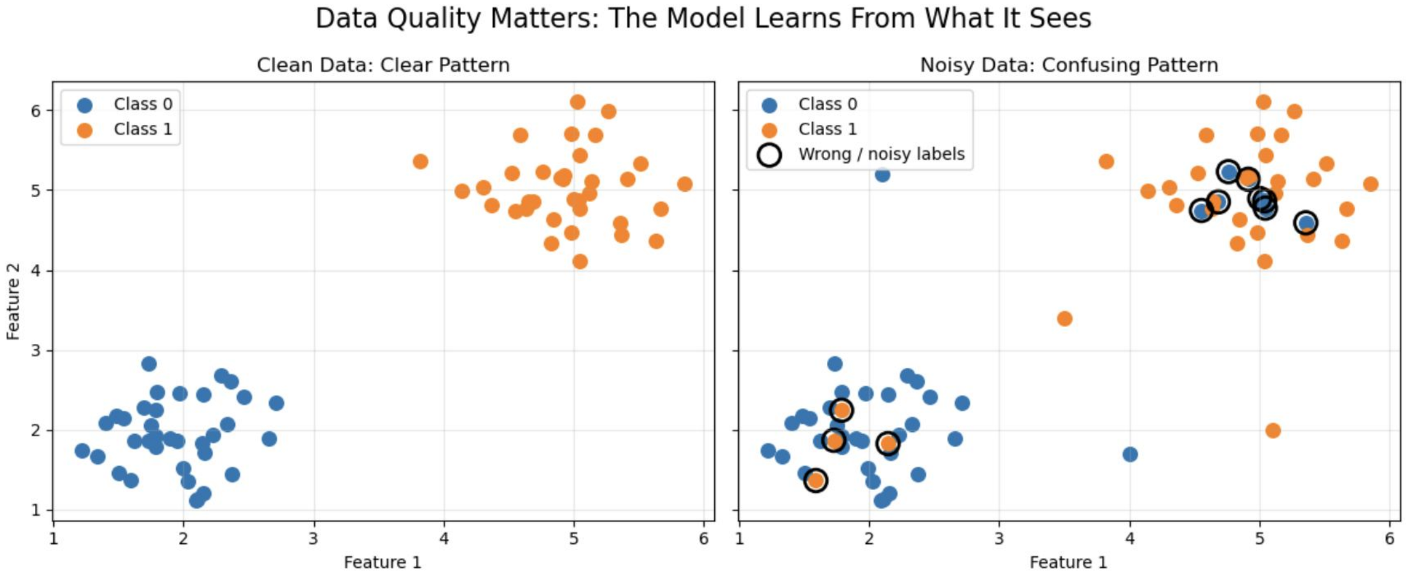
# Key Challenges in ML: Data Quality

These are all dogs.
No doubt.

# Key Challenges in ML: Data Quality

- Data is noisy or incorrect
- Model learns wrong patterns
- Output becomes unreliable

If the input is wrong, the model learns the wrong thing.



# Next

So far, we’ve seen different types of data and problems →

Linear regression & Logistic regression
ML Challenges & Regularization
Dimensionality Reduction & Topic Modelling & Principal Component Analysis
Neural networks
Bayes classification

# Mini Assignments

Clone the repo!

What you need:

- GitHub account
- Maybe a Google Colab account (not really necessary but could help)

TODOs:

- Exercise notebook